

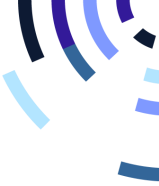
Probability, Statistics & Data Analysis

Lecture 2: Probability Theory

**Prediction is very difficult, especially
about the future**

Niels Bohr

Peer-Learning Study Group Trial



- **I'll suggest specific problems**
 - You will not know in advance which questions
 - I would like you to try as many of the exercises on your own as possible (for your benefit!)
- **Split into groups of 4-6**
 - Try to be a different group each week, it's good to work with new people and hear their understanding
 - We may try randomising if needed
- **As a group**
 - Discuss approaches to solve the problems
 - Pay attention to the non-numeric reflection and 'what can we conclude'-type questions.
 - If you're happy as a group, move on to the next problem
- **GTAs role will be**
 - To facilitate the conversations (to walk around the groups, asking how it's going)
 - If a particular problem brings up interesting differences between groups, or common difficulties, then expand to the whole room for a discussion on that question
 - Also – to answer and other questions on other questions, if the students have them
- *Please let the GTAs know whether this format was helpful*

2: Probability Theory

What we will cover

- Set Theory
- Axioms and Properties of Probability
- Counting
 - Product Rule
 - Permutations
 - Combinations
- Conditional Probability
- Total Probability and Bayes' Theorem
- Independence



Nomenclature

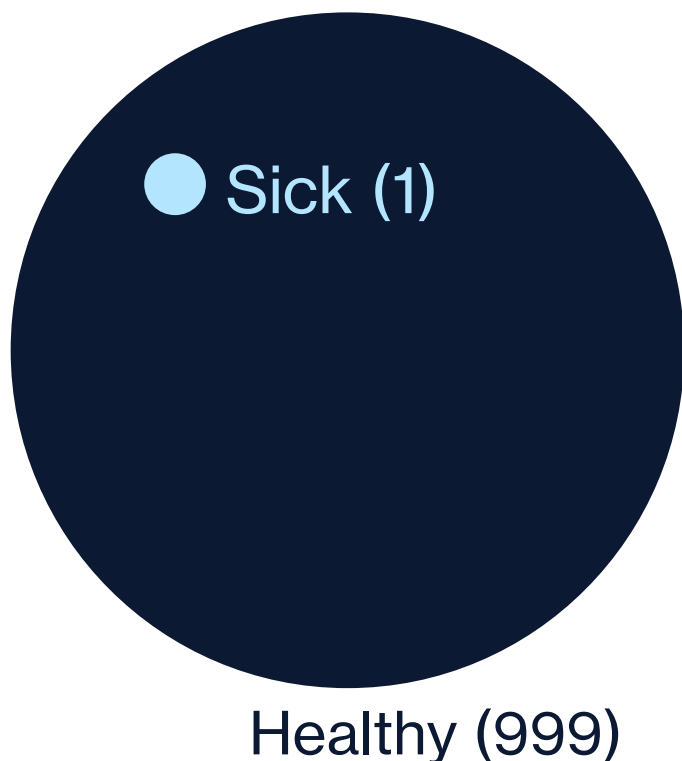
$\subset$	is a subset of
$\subseteq$	is a subset of or is equal to
$\cap$	intersection (AND)
$\cup$	union (OR)
$\setminus$	relative complement (AND NOT)
$\in$	is an element of
$\emptyset$	the empty set
$P(A) = x$	the probability of A occurring is x
$P(A B)$	the probability of A, if B has occurred



Disease Testing



- A transmittable disease affects 1 in 1000 people



- We need a diagnostic test to reliably identify who has the disease
- Our test of interest performs as follows

	Test Result	Probability
Healthy	$\xrightarrow{-}$ True negative	(98%)
	$\xrightarrow{+}$ False positive	(2%)
Sick	$\xrightarrow{-}$ False negative	(1%)
	$\xrightarrow{+}$ True positive	(99%)

If you get a positive result, what is the probability you are sick?

Disease Testing



We know the prevalence in the population already

Imperial College London

Example – Disease Testing



- A transmittable disease affects 1 in 1000 people



	Test Result		Probability
Healthy	-	True negative	(98%)
	+	False positive	(2%)
Sick	-	False negative	(1%)
	+	True positive	(99%)

If you get a positive result, what is the probability you are sick?

Statistics

2: Probability Theory

3

We know how the test performs already

- Prior knowledge of underlying distributions
- Hence probability rather than statistics
- We might use inferential statistics to ask
 - What is the prevalence of the disease in the population?
 - How well does the test perform?



Sett

Set Theory

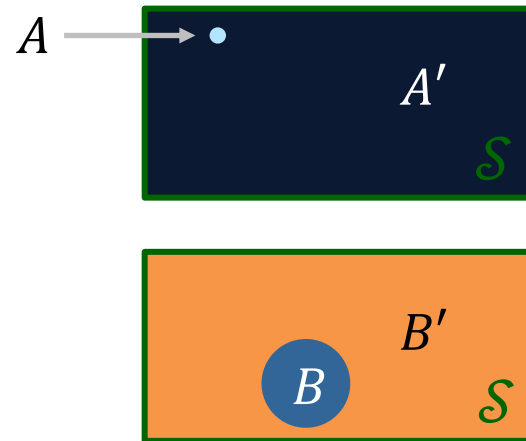


Sample Space



- Sample space $\mathcal{S}$ of an experiment is the set of all possible outcomes
- Event is a subset of outcomes within $\mathcal{S}$
- For example

- A = sick
- A' = healthy
- B = positive test result
- B' = negative test result



Healthy

→⁻ True negative

$A'B'$

→⁺ False positive

$A'B$

Sick

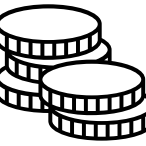
→⁻ False negative

AB'

→⁺ True positive

AB

Events



- Coin flip
 - Head: H
 - Tail: T
- Flip coin once

$$\{H, T\}$$

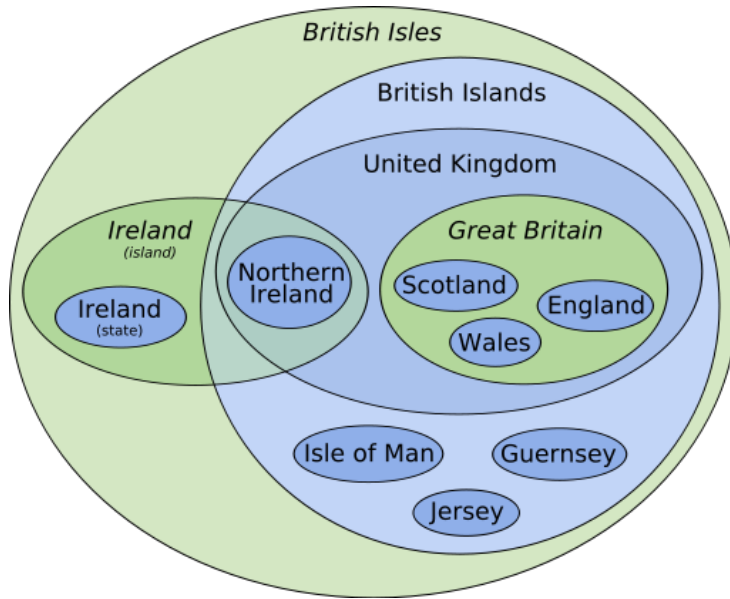
- Flip coin twice

$$\{HH, HT, TH, TT\}$$

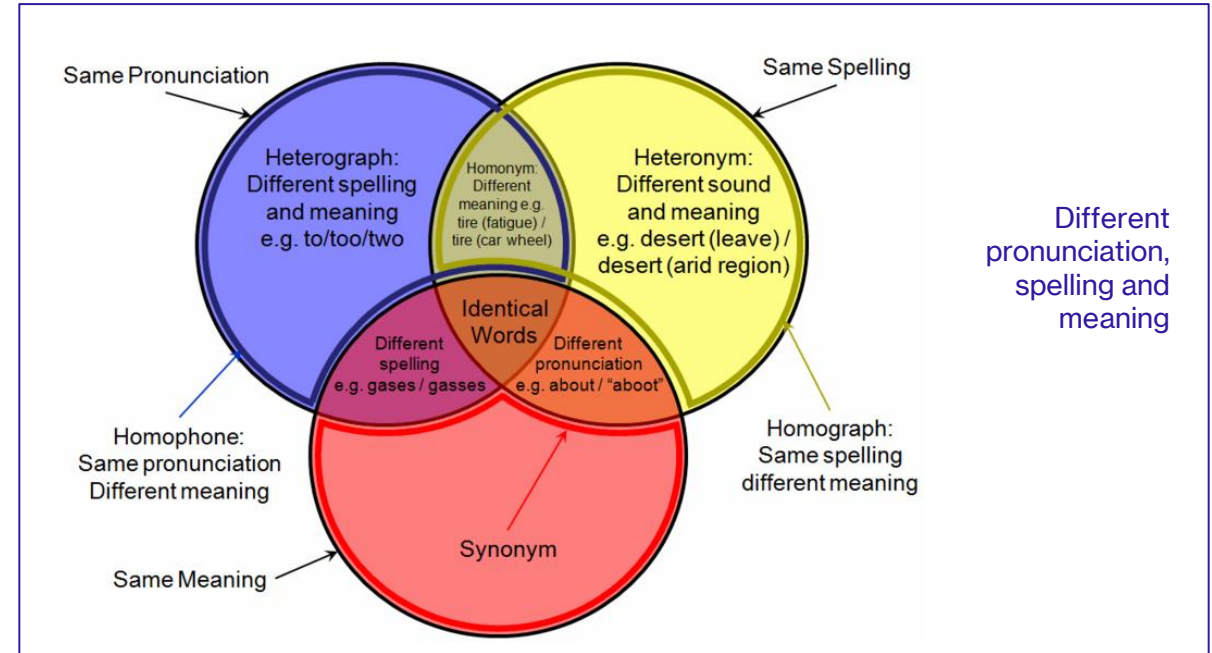
- An event is a collection (subset) of outcomes contained in the sample space
- Simple events – consists of exactly one outcome
 - Two heads – only one way for this to happen
 - $A = \{HH\}$
- Compound events – consists of more than one outcome
 - One head and one tail – two ways for this to happen
 - $B = \{TH, HT\}$

Venn Diagrams

British Isles



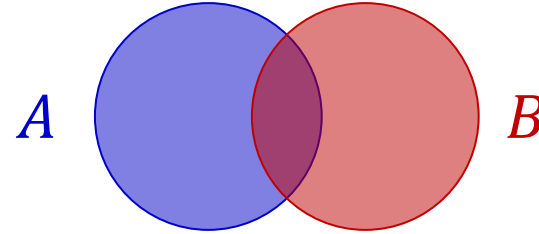
Word Categorisation



- British Isles is the 'set'
- The place being in the UK is an 'event'
- The place being in Wales is an 'event'

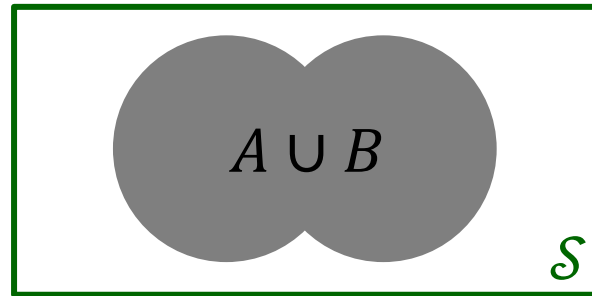
- The 'set' is the categorisation of a comparison between any two words (purple box)
- Synonym is an 'event'
 - – e.g. happy, content
- Homonym is an 'event'
 - – e.g. medium (size), medium (for cell culture), medium (communicating with spirits)

Set Theory



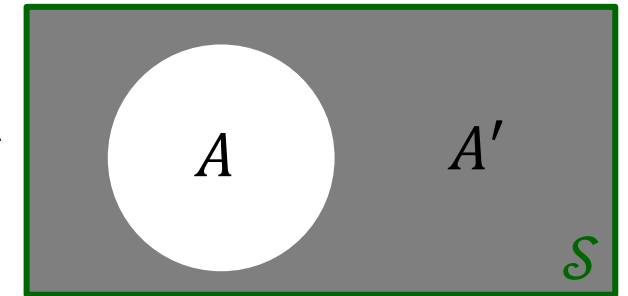
Union

Everything inside
A OR B



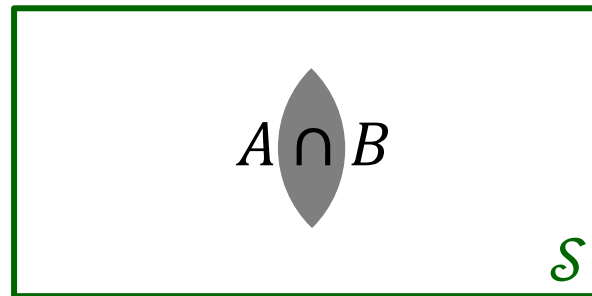
Complement

Everything not inside A
NOT(A)



Intersection

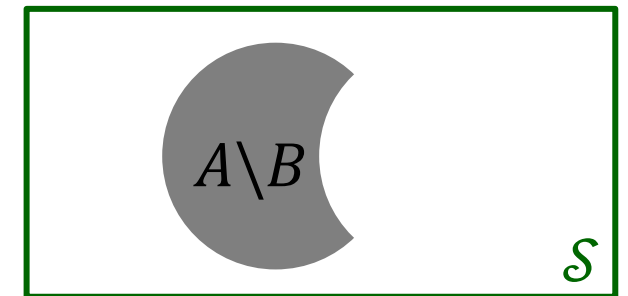
Everything inside
both **A AND B**



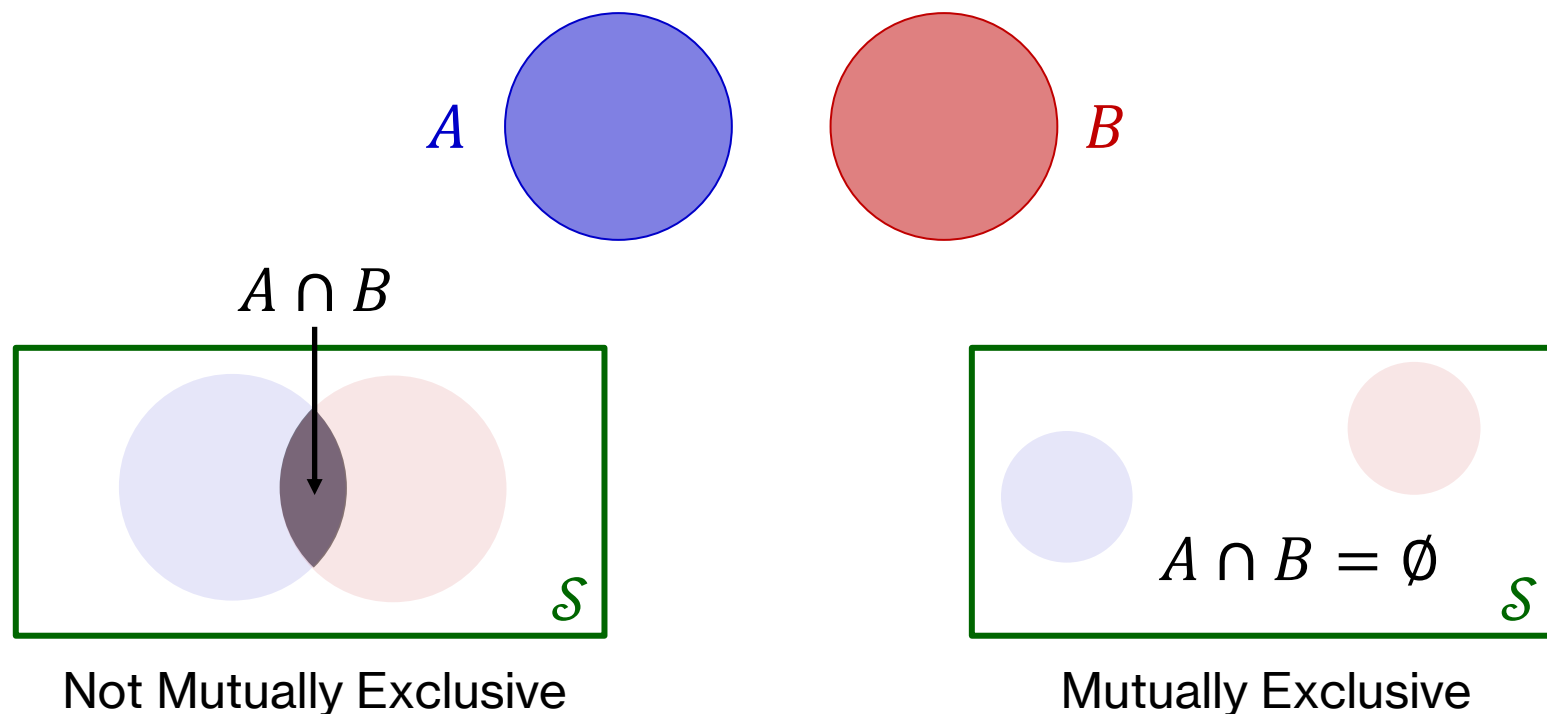
(∩ looks like A)

Relative Complement

Everything inside A
but not inside B
A AND NOT(B)



Mutual Exclusivity



- $\emptyset$ is the 'empty set'
- If A and B cannot occur together – mutually exclusive
- Also called 'disjoint'
 - E.g. Roll two dice. A =first dice yields 1, B =sum of two dice yields 8



Axioms of Probability

The building blocks of probability

Axioms (assumptions) of Probability

Axiom 1: for a given event, $A \subset \mathcal{S}$, the probability must be between 0 and 1

$$0 \leq P(A) \leq 1$$

Axiom 2: probability of the whole sample space is 1

$$P(\mathcal{S}) = 1$$

Axiom 3: for a collection of mutually exclusive events, $A_1, A_2, A_3, \dots$

$$P(A_1 \cup A_2 \cup A_3 \cup \dots) = \sum_{i=1}^{\infty} P(A_i)$$

U - union (OR)

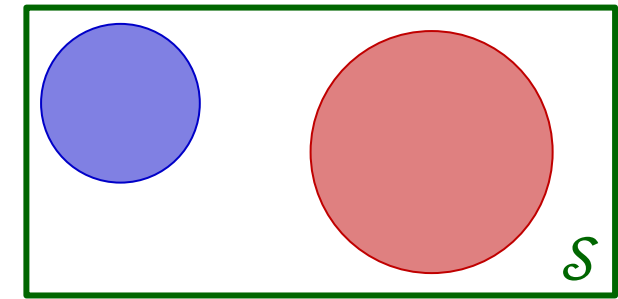


Roll two dice and define two events as specific outcomes for the sums of the two values:

$$A_1 = \{2, 3, 12\}$$

$$A_2 = \{7, 11\}$$

Find $P(A_1 \cup A_2)$



Axiom 3: for a collection of mutually exclusive events, $A_1, A_2, A_3, \dots$

$$P(A_1 \cup A_2 \cup A_3 \cup \dots) = \sum_{i=1}^{\infty} P(A_i)$$

The probability that any one of these events will occur is the sum of the probabilities of each event occurring

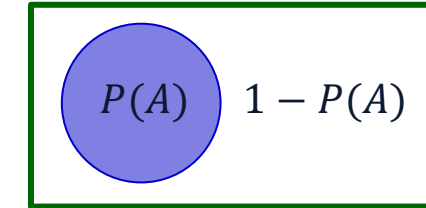
(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

Properties of Probability

1. For any event A , the probability it occurs is 1 minus the probability that it does not occur

$$P(A) = 1 - P(A')$$

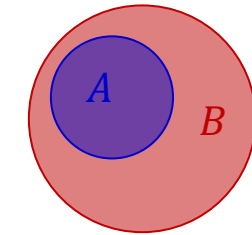
$$P(\mathcal{S}) = 1$$



2. Monotonicity

$$\text{If } A \subseteq B \text{ then } P(A) \leq P(B)$$

$\subseteq$ means 'is a subset of or is equal to'



3. Probability of the empty set is zero

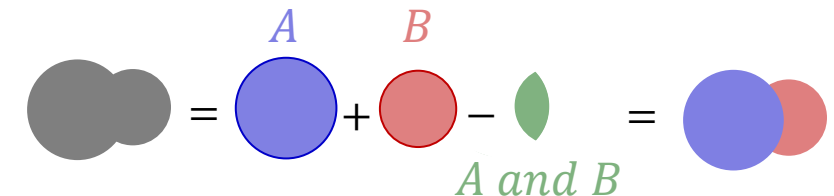
$$P(\emptyset) = 0$$

$$P(\mathcal{S})$$

'Something has to happen!'

4. Probability of A or B occurring

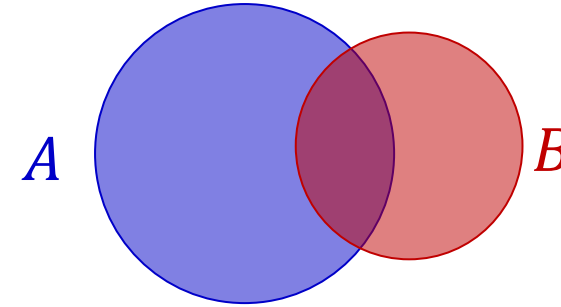
$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



Protein-Producing Bacteria



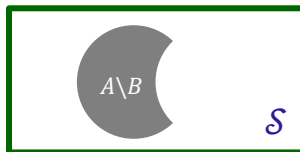
- A is production of protein 1
- B is production of protein 2
- From some experiments



$$\begin{aligned}P(A) &= 0.40 \\P(B) &= 0.65 \\P(A \cap B) &= 0.20\end{aligned}$$

- What is the probability that protein A is produced, but protein B is not?

**Relative
Complement**
Everything inside
A but not inside B
A and not(B)

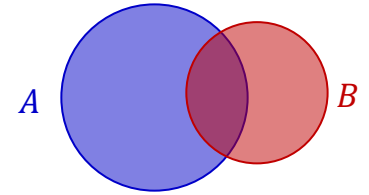
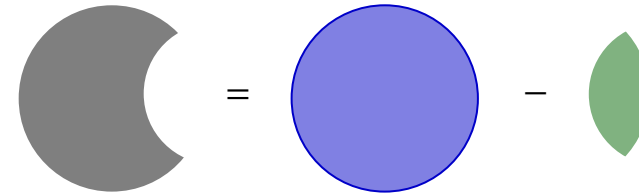


$$P(A \setminus B) = ?$$

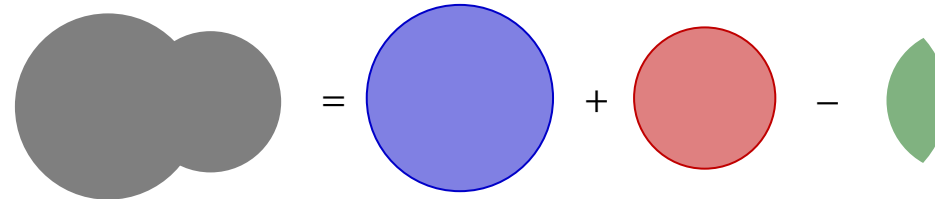
Protein-Producing Bacteria



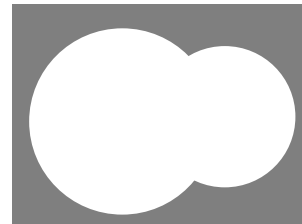
- What is the probability that protein A is produced, but protein B is not



- What is the probability that the bacteria produces at least one protein?



- Probability of no protein



$$\begin{aligned} P(A) &= 0.40 \\ P(B) &= 0.65 \\ P(A \cap B) &= 0.20 \end{aligned}$$



Counting

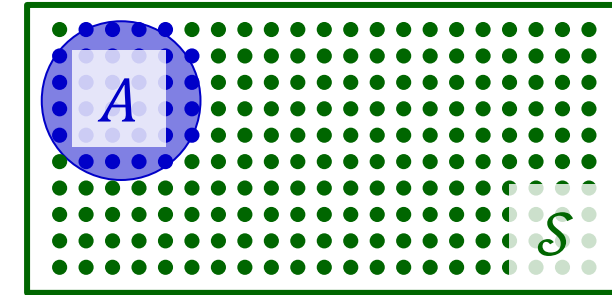
and ways to shortcut it

Frequentist Probability Definition

- Let $n(A)$ = the number of outcomes contained in event A
- Let n = the total number of outcomes

$$P(A) = \frac{n(A)}{n}$$

- Where $P(A)$ is the probability that a given outcome is contained in event A
- This is the 'relative frequency' understanding of probability
- Intuitive, but quite often impractical to manually count all of the outcomes
- Consider 3, 4, 5 or more dice....

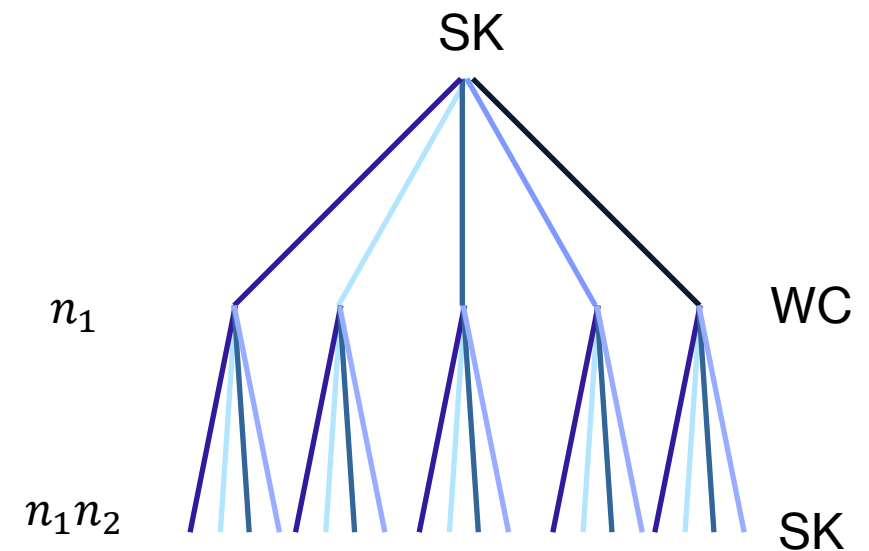


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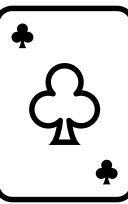


- Consider an event that consists of ordered pairs
 - I need to get from South Kensington to White City, and back again out of hours (no shuttle)
 - I can practically get from SK to WC $n_1 = 5$ ways
 - Cycle
 - Motorcycle
 - Bus+Walk
 - Tube + Walk
 - Shuttle
 - I can practically get back to SK from WC $n_2 = 4$ ways
 - Cycle
 - Motorcycle
 - Bus+Walk
 - Tube + Walk
- The total number of possible routes is
$$n_1 n_2 = 20$$
- Generally scales to k events:

$$n = n_1 \times n_2 \times \cdots \times n_k$$



Cards with replacement (independent)



- Consider a deck of cards (13 x 4 suits = 52 cards)
- What is the probability of getting two aces in a row...
- With replacements? (take a card, replace it)
 - There are 4 ways of getting an ace
 - Hence $P(A_1) = \frac{n(A_1)}{n_1} = \frac{4}{52} = 0.077$
 - Having replaced the card, the odds of getting another Ace are also $P(A_2) = 0.077$
 - Hence

$$P(A_1 \cap A_2) = \frac{n(A_1 \cap A_2)}{n_1 n_2} = \frac{4 \times 4}{52 \times 52} = \frac{16}{2704} = 0.0059$$

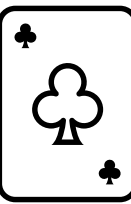
- Or equivalently

$$P(A_1 \cap A_2) = P(A_1)P(A_2) = 0.077 \times 0.077 = 0.0059$$

$$P(A) = \frac{n(A)}{n}$$

$$n = n_1 n_2$$

Cards without replacement (independent-ish)



- Without replacements (take a card, set it aside)
 - There are 4 ways of getting an ace on the first selection
 - Hence $P(A_1) = \frac{n(A_1)}{n_1} = \frac{4}{52} = 0.077$
 - There are 3 ways of getting an ace on the second selection, and 1 less card (this depends on the first selection)
 - Hence $P(A_2) = \frac{n(A_2)}{n_2} = \frac{n(A_1)-1}{n_1-1} = \frac{3}{51} = 0.059$
 - Hence

$$P(A_1 \cap A_2) = \frac{n(A_1 \cap A_2)}{n_1 n_2} = \frac{4 \times 3}{52 \times 51} = \frac{12}{2652} = 0.0045$$
 - Or equivalently

$$P(A) = \frac{n(A)}{n}$$

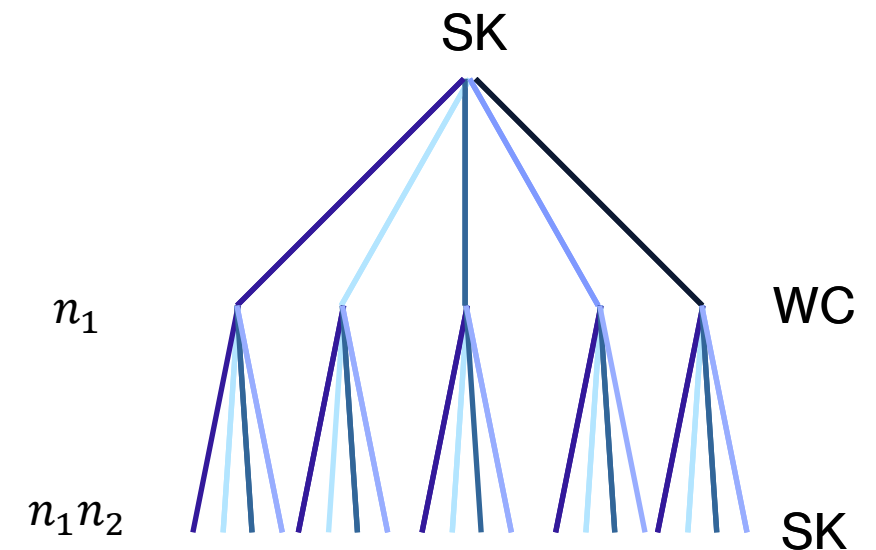
$$n = n_1 n_2$$

$$P(A_1 \cap A_2) = P(A_1)P(A_2) = 0.077 \times 0.059 = 0.0045$$

Even though A_2 strictly depends on A_1 , by adjusting the numbers in the middle step, we create a new, independent, scenario – pick an ace from a deck of 51 cards with three aces.



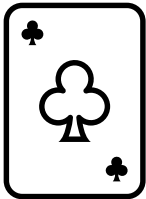
- For the product rule, each event was treated as independent of other events
 - The mode of transport you took one way doesn't affect the choice going the other way
- Let's consider that you don't like to use the same mode of transport twice in one day (due to boredom)
 - Formally, a given element can now only appear in one of the k positions
 - We have 4 fewer options



Counting rules: independent events

Permutations: dependent events, order matters

Cards Permutations



- Consider the card deck again
 - Select a card and set aside
 - How many possible outcomes can occur?
 - How many permutations of n objects for k selections, $P_{k,n}$?
 - $n = 52$ is the total number of events
 - $n_1 = 52$
 - $n_2 = 51$ $k = 3$
 - $n_3 = 50$ $k - 1 = 2$
 - $n_{\dots} = \dots$ $n_3 = 50$
 - $n_k = n - (k - 1)$

• Hence for k picks without replacement the number of outcomes are

$n = n_1 \times n_2 \times \dots \times n_k$ $P_{k,n} = n(n - 1)(n - 2) \dots (n - (k - 1))$

- If $k = n$, this equals $n!$
- For $k < n$ there are fewer options. In general:

$$P_{k,n} = \frac{n!}{(n - k)!}$$

Numerator – if you kept selecting until none were left
Denominator – the number of options left over when you're done

e.g. $n = 5, k = 3$

$$\begin{aligned} n! &= 5 \times 4 \times 3 \times 2 \times 1 \\ n - k &= 2 \\ (n - k)! &= 2 \times 1 \end{aligned}$$

$$P_{k,n} = \frac{5 \times 4 \times 3 \times 2 \times 1}{2 \times 1}$$

$$P_{k,n} = 5 \times 4 \times 3$$

$$P_{k,n} = 60$$

Doodle Poll - Permutations



- I have 4 PhD students I want to arrange weekly meetings with
- I have 10 possible time slots available
- How many different ways could this work out if order matters?
 - How many permutations are there?
- First student to respond can choose from 10
- Second student to respond can choose from 9
- Third student to respond can choose from 8
- Fourth student to respond can choose from 7

$$P_{k,n} = \frac{n!}{(n-k)!}$$

Numerator – if you kept selecting until none were left

Denominator – the number of options left over when you're done

```
>> factorial(10)/factorial(6)
ans =
    5040
```

Combinations

- Product Rule – Independent selections
- Permutations – Ordered, dependent selections
- Combinations – Unordered, dependent selections
- Combinations < Permutations
 - E.g. consider the set {A,B,C,D}
 - For events of size $k = 3$, there are 24 ($n!/(n - k)! = 4!/1!$) permutations
 - There are 6 ($=3!$) permutations of size three containing A, B, C
 - A,B,C – A,C,B – B,A,C – B,C,A – C,A,B – C,B,A
 - For any other combination of size 3 (e.g. {A, B, D}, {B, C, D}, {A, C, D}) there are also 3! permutations
 - If order doesn't matter, the 6 permutations yield a single combination
 - Generally, there are fewer combinations than permutations by a factor $k!$

$$C_{k,n} = \frac{P_{k,n}}{k!} = \frac{n!}{k! (n - k)!} = \binom{n}{k}$$

Doodle Poll - Combinations



- For the doodle poll, it doesn't actually matter what order I see the students in
- How many different ways are there that the meeting slots could be filled?

$$C_{k,n} = \frac{n!}{k! (n - k)!}$$

Numerator – if you kept selecting until none were left

Denominator – the number of options left over when you're done, times the number of permutations per combination

```
>> factorial(10)/factorial(6)/factorial(4)

ans =

    210
```

Doodle Poll Probability



- Now consider 6 of the slots are on Monday and 4 are on Wednesday
- What is the probability that I end up with 2 on Monday and 2 on Wednesday?
 - Assume all slots are equally likely to be selected
- Number of ways to get 2 on Monday

- Number of ways to get 2 on Wednesday

- Number of ways to get 2 on Monday and 2 on Wednesday

$$n = n_1 \times n_2 \times \cdots \times n_k$$

- As 210 possible combinations, the probability is given by



Conditional Probability

If something has already happened...

Conditional Probability

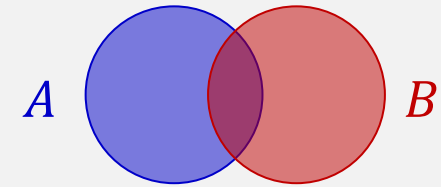
- Updating probability based on information that we have
- For example, if we already know that B has happened,
- The probability of A **given** B can be thought of like this:

$$P(A \cap B) = P(B) \cdot P(A|B)$$

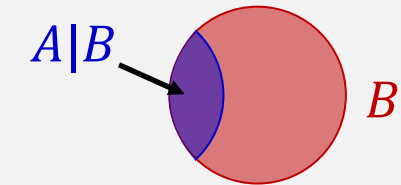
- (A and B happen) = (B happens) then (A happens given that B has happened)
- Re-arranging

Conditional
Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$



When B has happened, the sample space is reduced to the set B



The probability that A will then happen is

$A \cap B$

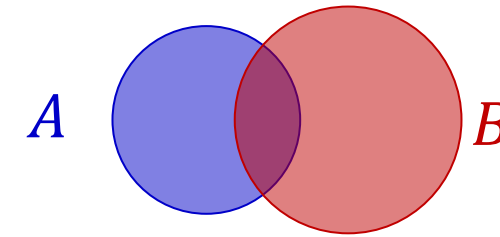


Protein-Producing Bacteria - Revisited



Bacteria Producing Proteins

- A is production of protein 1
- B is production of protein 2
- From some experiments



$$P(A) = 0.40$$

$$P(B) = 0.65$$

$$P(A \cap B) = 0.20$$

- If the bacteria has already produced protein 2, what is the probability that it will produce protein 1?

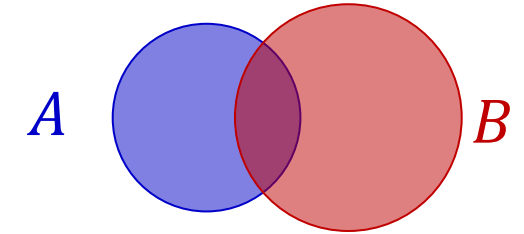
Conditional Protein-Producing Bacteria



- A is production of protein 1, B is production of protein 2
- If the bacteria has already produced protein 2, what is the probability that it will produce protein 1?
- If B , what is the probability A ?
- $P(A|B)$

Conditional
Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$



$$\begin{aligned}P(A) &= 0.40 \\P(B) &= 0.65 \\P(A \cap B) &= 0.20\end{aligned}$$

Conditional Probability (reversed)

- The conditional probability equation can be flipped (in terms of A and B)
- Note that the outcomes might be different
- The probability of **B given A** is

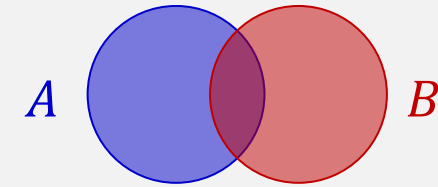
$$P(B \cap A) = P(A) \cdot P(B|A)$$

- (B and A happen) = (A happens) then (B happens given that A has happened)
- Re-arranging gives our conditional probability master equation

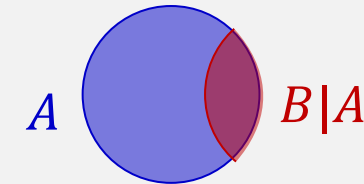
$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$

Conditional
Probability
Reversed

FYI



When A has happened, the sample space is reduced to the set A



The probability that B will then happen is
 $A \cap B$



Conditional Bacteria Reversed



- A is production of protein 1, B is production of protein 2
- If the bacteria has already produced protein 1, what is the probability that it will produce protein 2?
- If A , what is the probability B ?
- $P(B|A)$

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

- Not necessarily the same numerical value as $P(A|B)$!

$$\begin{aligned}P(A) &= 0.40 \\P(B) &= 0.65 \\P(A \cap B) &= 0.20\end{aligned}$$



Bayes' Theorem

using prior knowledge

Law of Total Probability

If events A_i are

- mutually exclusive (non-overlapping)
 - exhaustive ($\mathcal{S} = A_1 \cup A_2 \cup \dots \cup A_k$) (they define the whole set)
- Now consider an event B
 - $P(B) = P(B \cap A_1) + P(B \cap A_2) + \dots + P(B \cap A_k)$

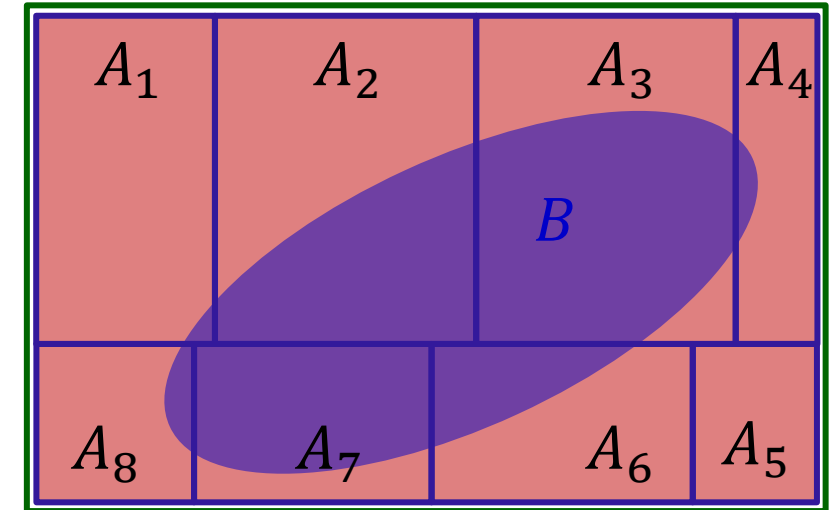
$$P(B \cap A_i) = P(A_i) \cdot P(B|A_i)$$

Then

$$P(B) = P(B|A_1)P(A_1) + \dots + P(B|A_k)P(A_k)$$

$$P(B) = \sum_{i=1}^k P(B|A_i)P(A_i)$$

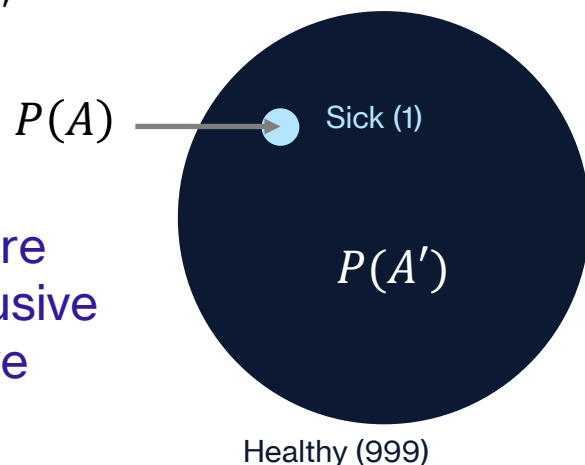
$\mathcal{S}$



Disease Testing



$$\mathcal{S} = A + A'$$



Note that A_i are mutually exclusive and exhaustive

Healthy	-	True negative	(98%)	$P(B' A') = 0.98$
	+	False positive	(2%)	$P(B A') = 0.02$
Sick	-	False negative	(1%)	$P(B' A) = 0.01$
	+	True positive	(99%)	$P(B A) = 0.99$

- A: individual is sick
- B: positive test result

related to condition
related to test result

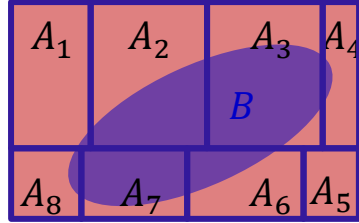
To interpret a test result, we need to ask

“If you get a positive result, what is the probability you are sick?”

What is $P(A|B)$?

- We know **result|condition**, but want **condition|result** – the reverse
- We have seen these are not necessarily the same

Bayes' Theorem Derivation



A_j is some subset of A_i
condition|result

$$P(A_j|B) = \frac{P(A_j \cap B)}{P(B)}$$

condition & result
result

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Conditional Probability

$$P(A_j|B) = \frac{P(B \cap A_j)}{P(B)}$$

result & condition
result

$$P(B \cap A_j) = P(A_j \cap B)$$

Intersection is reversible

$$P(A_j|B) = \frac{P(B|A_j)P(A_j)}{P(B)}$$

result|condition x condition
result

$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$

Conditional Probability Reversed

$$P(A_j|B) = \frac{P(B|A_j)P(A_j)}{\sum_{i=1}^k P(B|A_i)P(A_i)}$$

Bayes' Theorem



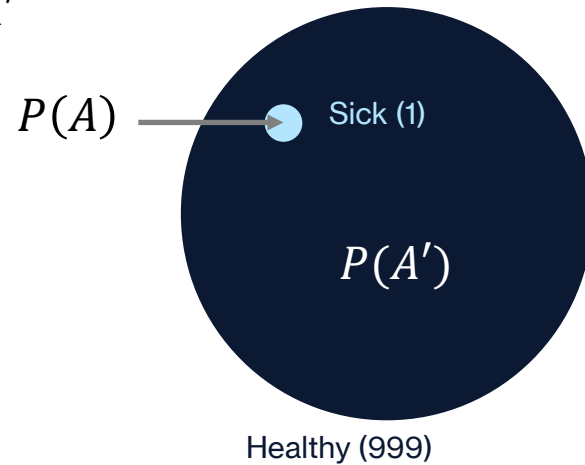
$$P(B) = \sum_{i=1}^k P(B|A_i)P(A_i)$$

Total Probability

Bayes Theorem - Rare Disease Testing



$$\mathcal{S} = A + A'$$



Healthy

- →

True negative

(98%)

$$P(B'|A') = 0.98$$

+ →

False positive

(2%)

$$P(B|A') = 0.02$$

Sick

- →

False negative

(1%)

$$P(B'|A) = 0.01$$

+ →

True positive

(99%)

$$P(B|A) = 0.99$$

What is $P(A|B)$? Sick, given positive test result

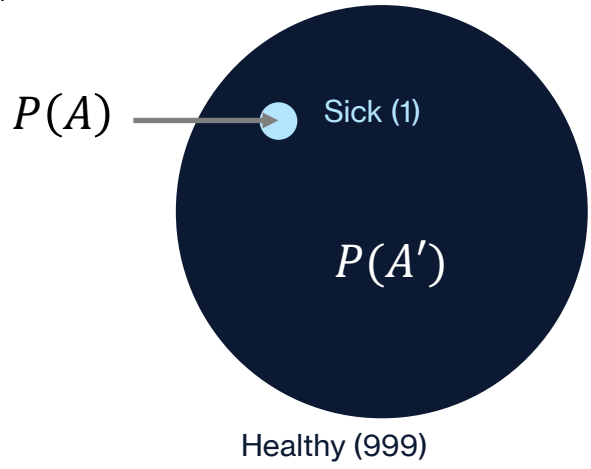
Bayes' Theorem

$$P(A_j|B) = \frac{P(B|A_j)P(A_j)}{\sum_{i=1}^k P(B|A_i)P(A_i)}$$

Rare Disease Testing - Intuition



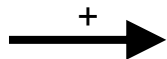
$S = A + A'$



Healthy



True negative (98%) $P(B'|A') = 0.98$

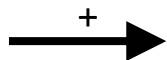


False positive (2%) $P(B|A') = 0.02$

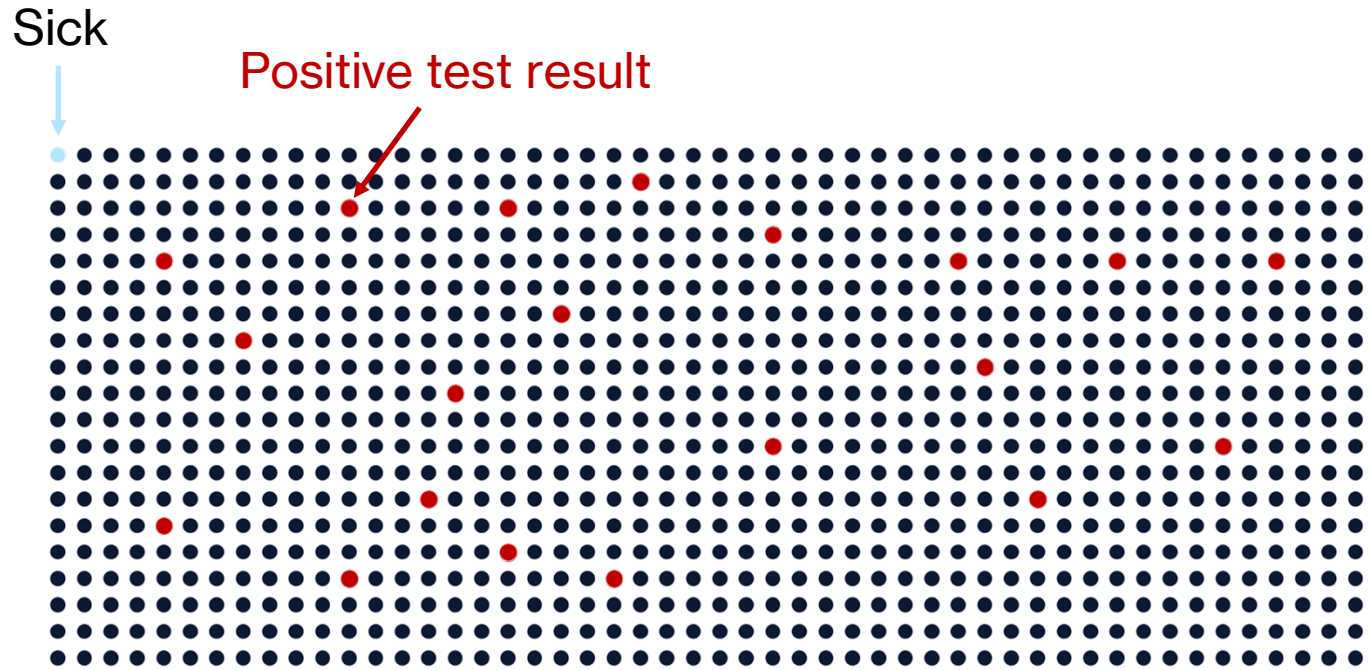
Sick



False negative (1%) $P(B'|A) = 0.01$



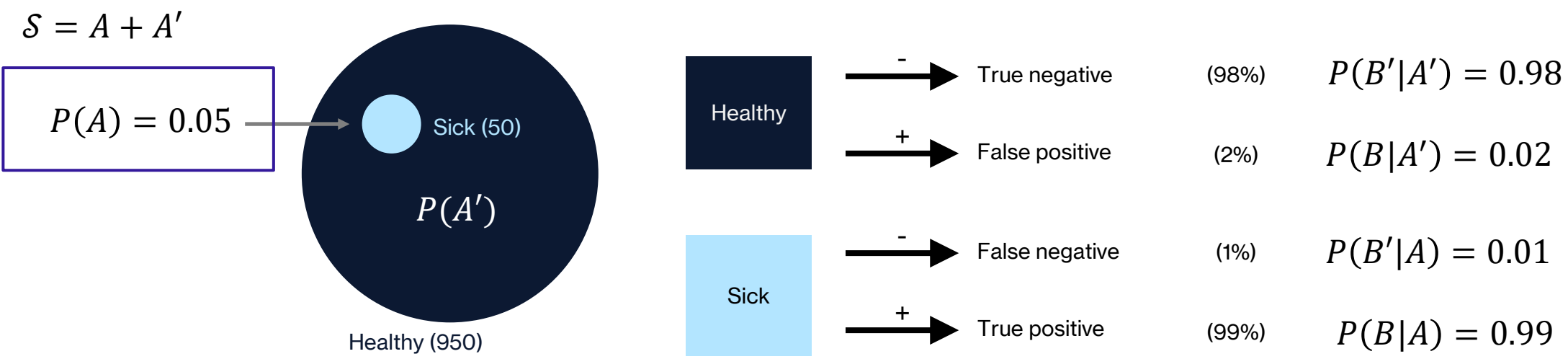
True positive (99%) $P(B|A) = 0.99$



If you get a positive result, what is the probability you are sick?

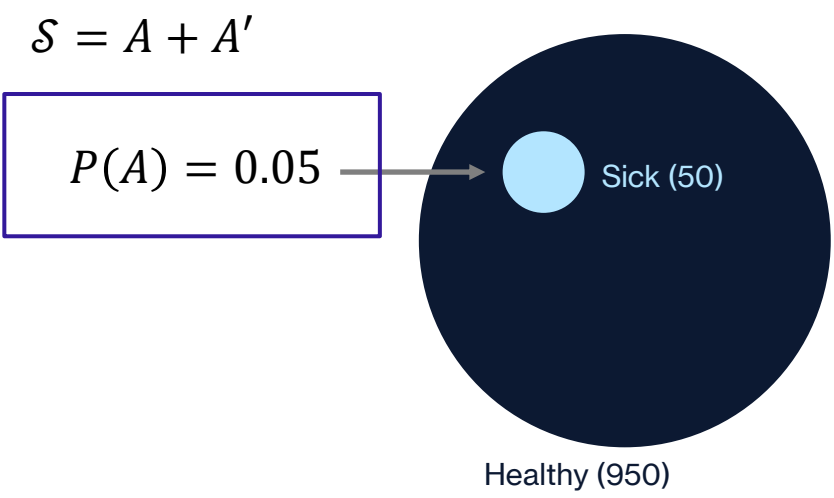
- **Approx. 21 positive tests**
 - ~20 false positive 
 - ~1 true positive 
- $P \sim 1/20 = 0.05$

Bayes Theorem - Less Rare Disease Testing



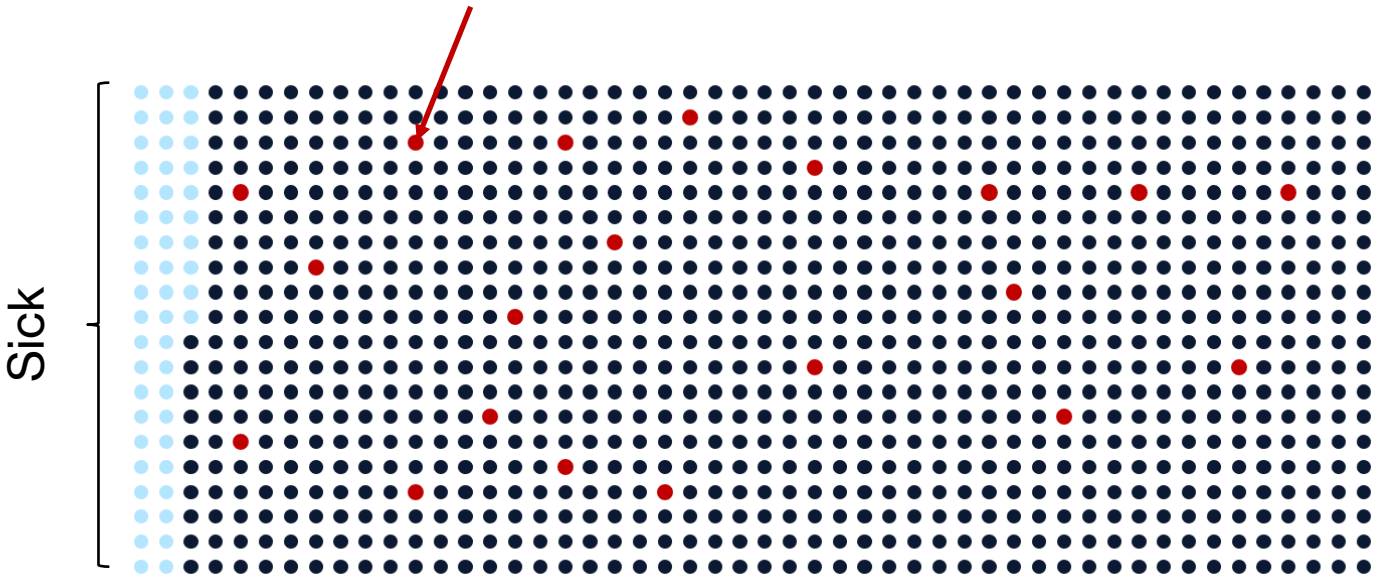
What is $P(A|B)$ when disease is more prevalent?

Less Rare Disease Testing - Intuition



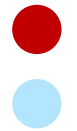
Healthy	-	True negative	(98%)	$P(B' A') = 0.98$
	+	False positive	(2%)	$P(B A') = 0.02$
Sick	-	False negative	(1%)	$P(B' A) = 0.01$
	+	True positive	(99%)	$P(B A) = 0.99$

Positive test result



If you get a positive result, what is the probability you are sick?

- **Approx. 70 positive tests**
 - ~20 false positive
 - ~50 true positive
- $P \sim 50/70 = 0.72$



Bayes Theorem - Understanding

A: individual is sick
B: positive test result

Probability of having the disease given a positive test result

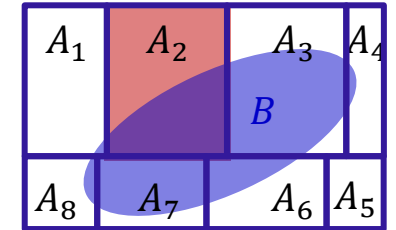
Probability of a true positive

Probability of having the disease

$$P(A_j|B) = \frac{P(B|A_j)P(A_j)}{\sum_{i=1}^k P(B|A_i)P(A_i)}$$

$$P(A|B) = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|A')P(A')}$$

Probability of a positive test result (irrespective of the disease)



- When prevalence increased
 - Likelihood of getting a positive test, $P(B)$, increased (~ 3 fold)
 - Probability of having disease, $P(A)$, increased a lot more (50 fold)
- Bayes Theorem is foundation for Bayesian Statistics
 - Beyond the scope of this course

If you are interested, read [this blog post](#)



Independence

An important consideration in experiments

Two-Way Independence

Two Events

Independent – A is unaffected by B

or equivalently

$$P(A|B) = P(A)$$

$$P(A \cap B) = P(A)P(B)$$

Conditional probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

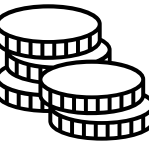
- Dependent – A is affected by B

$$P(A|B) \neq P(A)$$

Examples –

- Select an individual from the class
 - Height > 180 cm and weight < 70 kg (dependent)
- Select an individual from this class and one from another
 - First student is European and second student is European (independent)

Coin Flips I



Tossing a coin

- A: Head on the second throw
- B: Tail on the first throw

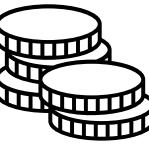
Independent if

$$P(A|B) = P(A)$$

or

$$P(A \cap B) = P(A)P(B)$$

Coin Flips II



Tossing a coin twice

- A: Head on the first throw
- B: Two heads

Independent if

$$P(A|B) = P(A)$$

or

$$P(A \cap B) = P(A)P(B)$$

Conditional probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Multiple-Way Independence



Three Events

- Mutually independent if all of the below are true

$$P(A \cap B \cap C) = P(A)P(B)P(C)$$

$$P(A \cap B) = P(A)P(B)$$

$$P(B \cap C) = P(B)P(C)$$

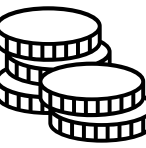
$$P(A \cap C) = P(A)P(C)$$

i.e. no event affects any other event

- Three dice
- Three coin tosses
- Three individuals that have never crossed paths taking tests for a transmissible disease

Generalisable to n events

Mutually Exclusive vs Independent



Mutually Exclusive: A and B cannot both happen (think simultaneously)

- One coin toss
 - A: a head is thrown
 - B: a tail is thrown



Independent: A does not effect B and vice versa (think sequentially)

- Two coin tosses
 - A: a head is thrown on the first coin
 - B: a tail is thrown on the second coin



then

2: Probability Theory

What we covered

- Set Theory
- Axioms and Properties of Probability
- Counting
 - Product Rule
 - Permutations
 - Combinations
- Conditional Probability
- Total Probability and Bayes' Theorem
- Independence

Next Up

Lecture 3: Discrete Random Variables





Appendix

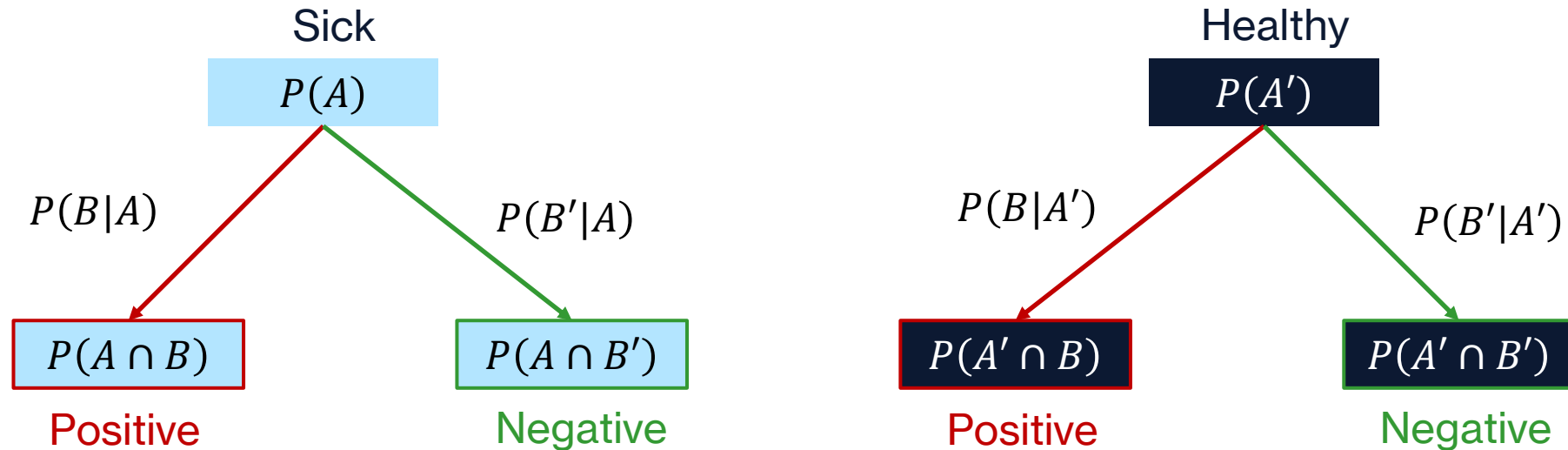
A second test for a rare disease

Is it worth repeating the test for a rare disease?



- When a disease is rare, even a small false positive rate can result in a low probability of having the disease after getting a positive test
- It is therefore common practice, following a positive test, to take a second test, with the following conclusions
 - If the second test was negative, we'd probably assume we're ok
 - If the second tests was positive, we'd assume that we have the disease
- Formally, we would like to know the probability of having the disease, given two positive tests
- First, I want to introduce another way to visualise Bayes Theorem in general, which I think might help: drawing a Tree Diagram of the possible outcomes

Tree Diagram Approach



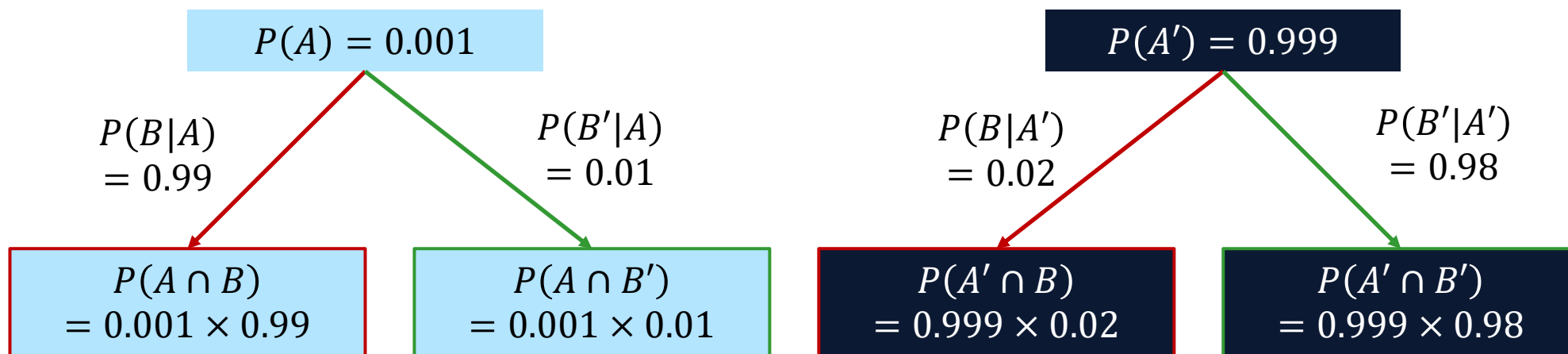
1. Start with the two options for A, then give the two possible test results
2. Use $P(B|A)P(A) = P(B \cap A) = P(A \cap B)$ (Conditional Probability reversed)
3. Each row now gives the probabilities of the possible outcomes
4. Ultimately, we want (from Conditional Probability)

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(\text{Sick} | \text{Positive}) = P(\text{Sick} \cap \text{Positive}) / P(\text{Positive})$$

4. For the numerator, we just have one outcome
5. For the denominator, we add the probabilities of all outcomes that include a positive test result (red boxes)

Tree Diagram With Numbers (1 test)



Plugging in the numbers:

$P(A \cap B) = 0.001 \times 0.99 = 0.00099$ is the probability of sick and positive test

$P(B) = P(A \cap B) + P(A' \cap B) = 0.00099 + 0.01998 = 0.02097$ is the probability of a positive test

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.00099}{0.02097} = 0.047$$

What about taking a second test?

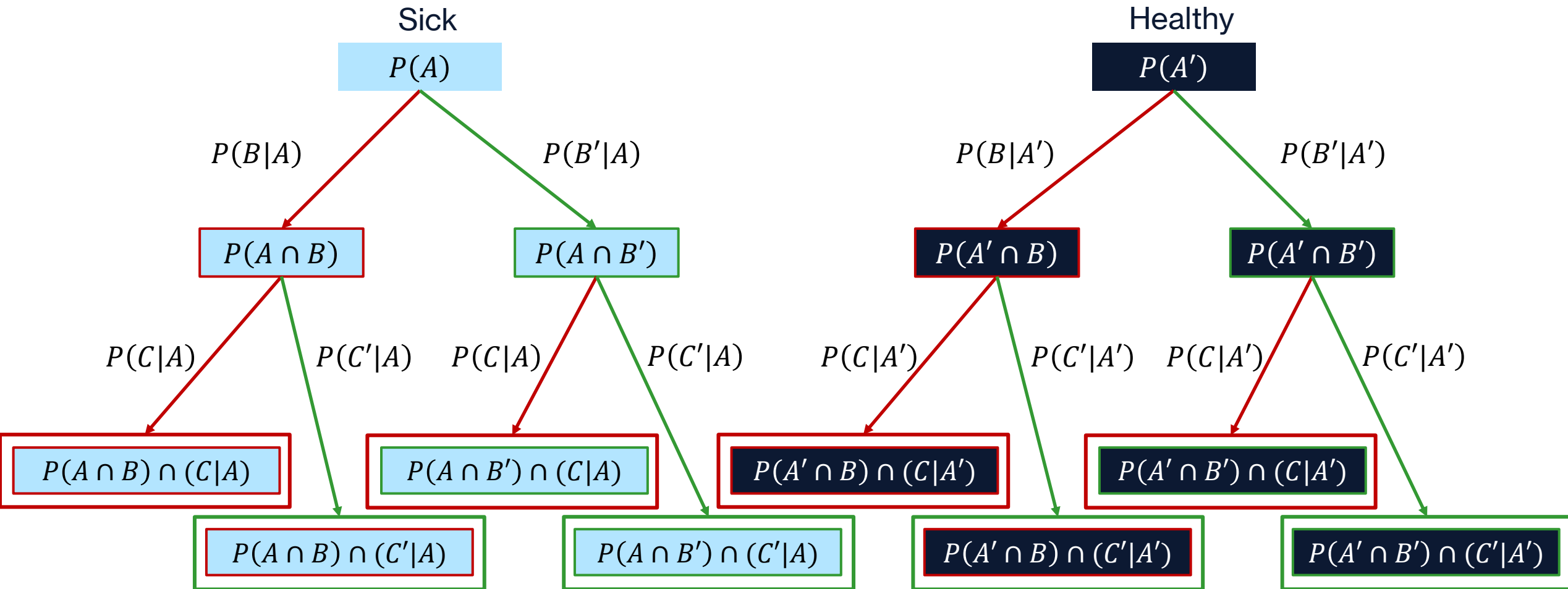


- Now we follow the same process to ask ‘what is the probability of being sick if you get two positive tests
- Intuitively: we assume that the tests are independent
- Setting our as we did before:

	1 st Test Result	Probability	2 nd Test Result	Probability
Healthy	- True negative	98%	-	~96%
	+ False positive	2%	+	~2%
			-	~2%
			+	~0.04%
Sick	- False negative	1%	-	~0.01%
	+ True positive	99%	+	~1%
			-	~1%
			+	~98%

Tree Diagram – Second Test

- Letting C be the second test



- The tree diagram is a bit busy, but we basically just need the two cases with double red outlines (2 positive tests)

Tree Diagram Approach

- What we want is

$$P(\text{Sick} \mid 2 \text{ Positive}) = P(\text{Sick} \ \& \ 2 \text{ Positive}) / P(2 \text{ Positive})$$

- 2 positive is represented by two red outlines in the diagram, from which we can see we want

$$P(A \mid (B \cap C)) = \frac{P(A \cap B) \cap (C \mid A)}{P(A \cap B) \cap (C \mid A) + P(A' \cap B) \cap (C \mid A')}$$

Denominator and 1st term of numerator

$$P(A \cap B) \cap (C \mid A)$$

- or reversing the conditional probability definition

$$P(A)P(B \mid A) \cap (C \mid A)$$

- Given that $C \mid A$ and $B \mid A$ are independent, we can therefore write

$$P(A)P(B \mid A)P(C \mid A)$$

- Which given that $P(B \mid A) = P(C \mid A)$, becomes

$$P(A)P(B \mid A)^2$$

2nd term of numerator

- Following the same logic, this reduces to

$$P(A')P(B \mid A')^2$$

Tree Diagram Approach

Hence we have

$$P(A|(B \cap C)) = \frac{P(A)P(B|A)^2}{P(A)P(B|A)^2 + P(A')P(B|A')^2}$$

From which we can calculate

$$P(A|(B \cap C)) = \frac{0.99^2 \times 0.001}{0.99^2 \times 0.001 + 0.02^2 \times 0.999} = \frac{0.0009801}{0.0009801 + 0.0003996} = 0.047$$

$$P(A|(B \cap C)) = \frac{0.9801}{0.9801 + 0.3996} = 0.714$$

Quick Approach

A more straightforward approach here is just to replace B with BB (two positive tests)

$$P(A|BB) = \frac{P(BB|A)P(A)}{P(BB|A)P(A) + P(BB|A')P(A')}$$

We just need $P(BB|A)$ and $P(BB|A')$

As the two tests are independent

- $P(BB|A) = P(B|A)P(B|A) = P(B|A)^2$
- $P(BB|A') = P(B|A')P(B|A') = P(B|A')^2$
- Hence

$$P(A|BB) = \frac{P(B|A)^2 P(A)}{P(B|A)^2 P(A) + P(B|A')^2 P(A')}$$

- Which is the same as our earlier solution on the previous page

Intuition

- If you get a positive result on the first test, what is the probability you are sick?
- Approx. 21 positive tests
 - ~1 true positive
 - ~20 false positive
- Let's imagine we started with 50000 instead of 1000
- This would now be (by scaling)
 - ~50 true positives
 - ~1000 false positives
- Disease prevalence in the subpopulation with a positive first test is ~0.047 (from initial answer)
 - Pretty close to our less rare disease (which yielded ~0.7)
- The first test selected for a subpopulation with a higher disease prevalence
 - So the second test did a better job
 - Although 0.7 is still not that good!